

Drone Payload Detection - Data Collection Report

Data Collection Summary

Equipment Used

- 4K Camera
- Raspberry Pi Camera

Test Objects

1. Transparent Box
2. Thermocol Box
3. Two Additional Random Boxes (used in combination tests)

Detailed Data Collection Table

Test No.	Environment	Surface Type	Lighting Condition	Test Configuration	Camera Type	Data Captured
1	Indoor	Table	Room Lighting	Transparent box only (all angles)	4K Camera	Videos + Photos
2	Indoor	Table	Room Lighting	Thermocol box only (all angles)	4K Camera	Videos + Photos
3	Indoor	Table	Room Lighting	Both boxes together (all angles)	4K Camera	Videos + Photos
4	Indoor	Table	Room Lighting	All 4 boxes together (all angles)	4K Camera	Videos + Photos
5	Indoor	Grass Floor Mat	Room Lighting	Transparent box only (all angles)	4K Camera	Videos + Photos
6	Indoor	Grass Floor Mat	Room Lighting	Thermocol box only (all angles)	4K Camera	Videos + Photos

Test No.	Environment	Surface Type	Lighting Condition	Test Configuration	Camera Type	Data Captured
7	Indoor	Grass Floor Mat	Room Lighting	Both boxes together (all angles)	4K Camera	Videos + Photos
8	Indoor	Grass Floor Mat	Room Lighting	All 4 boxes together (all angles)	4K Camera	Videos + Photos
9	Indoor	Grass Floor Mat	Low Light (lights off)	Transparent box only (all angles)	4K Camera	Videos + Photos
10	Indoor	Grass Floor Mat	Low Light (lights off)	Thermocol box only (all angles)	4K Camera	Videos + Photos
11	Indoor	Grass Floor Mat	Low Light (lights off)	Both boxes together (all angles)	4K Camera	Videos + Photos
12	Indoor	Grass Floor Mat	Low Light (lights off)	All 4 boxes together (all angles)	4K Camera	Videos + Photos
13	Outdoor	Long Grass (Ground)	Natural Light	Transparent box only (all angles)	4K Camera + Raspberry Pi Cam	Videos
14	Outdoor	Long Grass (Ground)	Natural Light	Thermocol box only (all angles)	4K Camera + Raspberry Pi Cam	Videos
15	Outdoor	Long Grass (Ground)	Natural Light	Both boxes together (all angles)	4K Camera + Raspberry Pi Cam	Videos
16	Outdoor	Concrete Floor	Natural Light	Transparent box only (all angles)	4K Camera + Raspberry Pi Cam	Videos
17	Outdoor	Concrete Floor	Natural Light	Thermocol box only (all angles)	4K Camera + Raspberry Pi Cam	Videos
18	Outdoor	Concrete Floor	Natural Light	Both boxes together (all angles)	4K Camera + Raspberry Pi Cam	Videos

Test Conditions Summary

Environment Types

- **Indoor:** Controlled environment with table and grass floor mat
- **Outdoor:** Natural environment with long grass and concrete surfaces

Surface Types Tested

1. Table (smooth surface)
2. Grass Floor Mat (textured indoor surface)
3. Long Grass - Ground (natural outdoor terrain)
4. Concrete Floor (hard outdoor surface)

Lighting Conditions

1. Room Lighting (standard indoor illumination)
2. Low Light (lights turned off)
3. Natural Light (outdoor conditions)

Box Configurations

1. Individual boxes (transparent and thermocol separately)
2. Both primary boxes together
3. All four boxes together (including two random boxes)